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makeTwin: A reference architecture for digital twin software platform



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Abstract Digital twin is currently undergoing a significant transformation from the conceptual and theoretical research phase to the implementation and application phase. However, a universally adaptable research and development platform for digital twin is critically needed to meet the development requirements. Specifically, a publicly accessible simulation, testing, and validation platform which can support digital twin model building, data processing, algorithm design, configuration, etc., is urgently required for researchers. Furthermore, for developers from the industry, a low-code development platform that can offer a customizable suite of functions such as model creation, data management, protocol configuration, and visualization is urgently needed. Meanwhile, for enterprise users, there is a lack of an application management platform that can be configured and migrated for various application scenarios, functions, and modes. Therefore, based on the system research of digital twin theories and key technologies by the authors (such as the five-dimension digital twin model, digital twin modeling and digital twin data theory, digital twin standards, and so

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on), a digital twin software platform reference architecture, namely makeTwin, is proposed and designed, as well as its ten core functions. The workflow of the makeTwin and the interaction mechanisms among its core functions are described. Finally, a digital twin application system for a chemical fiber textile shop floor (CFTS) which was developed according to makeTwin, is introduced, which validates the proposed reference architecture.

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1. Introduction

Digital Twin can depict, emulate, forecast, optimize, and control the physical space through real-time connectivity, mapping, analysis, and interaction. Powered by multidimensional models and integrated data, digital twin removes the limitations of various physical conditions. With the advantages of model, data and service, digital twin offers superior solutions for enhancing quality, increasing efficiency, cutting costs, mitigating losses, ensuring safety, and conserving energy, etc. Therefore, digital twin enables maximal optimization of processes and value chains of the physical system. Moreover, the application of digital twin in diverse industries can effectively expedite their digitalization, networking, and intelligence development processes. These effort leads to continuous improvement of services, shorten response time to demand, optimization of processes, as well as upgrading of industries. Therefore, digital twin has become a powerful driving force for industrial intelligent transformation^{1–2}.

The concept of digital twin originated from NASA's Apollo moon landing program. Dr. Michael Grieves defined a three-dimensional digital twin model, which consists of three parts, namely physical products, virtual products, and connected data.³ In 2012, NASA first applied digital twin to aircraft structure maintenance. It was defined as the integrated multi-physics, multi-scale and accurate simulation of a complex product by using the best available physical models, sensors, etc., to mirror the life of its corresponding twin.⁴ Since then, digital twin has received increasing attention and research from academia and industry worldwide. In recent years, major global economies have begun to layout digital twin strategic plans, such as the Digital Twin Consortium in United States,⁵ the National Digital Twin Programme in the UK,⁶ Digital Twin City Plan in France,⁷ Industrial Digital Twin Association in Germany,⁸ and Digital Twin Computing initiative in Japan⁹.

The current research on digital twin is progressively delving into their fundamental theory and key common technologies. Basic theories, including the five-dimensional digital twin model,¹⁰ theories for digital twin modelling,¹¹ digital twin data,¹² and digital twin interaction,^{13–14} as well as digital twin application technology and tool system,¹⁵ have undergone in-depth research and analysis. Meanwhile, academia and industry have explored digital twin solutions in product design,¹⁶ intelligent shop floor and production line,^{17–19} complex task scheduling optimization,^{20–21} equipment fault diagnosis and prediction,²² equipment health management, and life prediction.²³ With these researches a preliminary standard system was built.²⁴ Correspondingly, digital twin has been applied

and explored in various fields such as aerospace,^{25–26} satellite,^{27–28} rail transportation,^{29–30} shipping,^{31–32} automobile manufacturing,³³ healthcare,^{34–35} power system,^{36–37} and urban management^{38–39}.

However, the lack of a universally applicable digital twin platform for research and development hinders digital twin's further development both in theoretical research and industrial application. Specifically, when researching the digital twin's fundamental theory and key technologies, a comprehensive platform is needed to perform simulation analysis and test verification of digital twin models, data, algorithms, interactions, and other related components. When implementing specific digital twin project development, developers require a development platform to support rapid configuration and invocation of low-code development for digital twin model generation, interactive configuration, data processing, algorithm selection, and visualization development. When utilizing digital twin application services, enterprise users expect to be able to rapidly configure, switch, and migrate deployed application software based on varying using scenarios and product models. To rapidly fulfill the digital twin market requirements, world-renowned software vendors are developing customized digital twin solutions for various industries and have released software platforms related to digital twin.¹⁵ However, it is noteworthy that most platforms only focus on limited dimensions or functionalities. There remains a lack of commercial digital twin platforms which can fulfill comprehensive functional requirements. From an openness and compatibility standpoint, enabling tools/platforms in the relevant industry tend to form closed software ecosystems around their respective products. Thus, it is difficult to interact and integrate models and data among different tools and platforms, leading to poor collaboration ability and compatibility.

To address the above challenges and provide a universally applicable digital twin industrial software platform architecture for academic researchers, project developers and enterprise users, the makeTwin, a digital twin software platform reference architecture, is proposed and designed in this paper. It is comprised of ten core functional modules, which are the *twinModelBuilder*, the *twinDataProcessor*, the *twinAlgBuilder*, the *twinIoTConnector*, the *twinInteractor*, the *twinSimulator*, the *twinVisualization*, the *twinLibrary*, the *twinSceneTemplate*, and the *twinAppDeployer*. The makeTwin aims to provide academic researchers with an open platform for digital twin theoretical innovation research and verification. Besides, it also aims to offer a general configurable and reusable digital twin development platform for project developers. Furthermore, it is also designed to supply convenient and reliable management for digital twin services for enterprise users.

The remainder of this paper is organized as follows. Different users' demands for the digital twin tools or software plat-

form are analyzed in Section II, and the current development status of these tools are reviewed. Section III gives the design concept of the makeTwin, and introduces its ten core functional modules. Section IV presents the workflow of the makeTwin as well as its interaction mechanisms. Section V introduces the digital twin system for a chemical fiber textile shop-floor (CFTS) which is developed according to the proposed makeTwin reference architecture. Section VI concludes this work and points out the future works.

2. Demands and definitions of digital twin software platform

Industrial software is an important carrier for the implementation of the digital twin. Researchers, developers, and enterprise users all rely on digital twin software platforms to support the development of digital twin application services. On the one hand, digital twin software platforms are expected to have features such as configurability, callability, modifiability, and extendibility for digital twin models, data, algorithms, IoT connectivity, interaction, simulation, and visualization. On the other hand, digital twin software platforms should be comprehensive, universal, and developmental to meet the different needs of researchers, developers, and enterprise users. Therefore, a detailed survey is conducted, and the different users' demands for digital twin software platforms are summarized, as shown in Fig. 1 and Table 1.

(1) Digital twin model

Various users expect digital twin industrial software that can meet different functions when conducting research and development on digital twin models. Such functions are model

import, model creation, model conversion, model lightweight, model assembly, model fusion, model validation, and model correction. The specific demands are as follows.

- **Model import.** It should support a diverse range of geometry models in multiple formats. Software with this capability includes, but is not limited to, TwinBuilder,⁴⁰ AWS IoT TwinMaker,⁴¹ Azure Digital Twins,⁴² and 3DEXPERIENCE.⁴³ Additionally, it should have the ability of importing various types of mechanism models, including physics, behavior, and rule models, as exemplified by 3DEXPERIENCE⁴³ and TwinBuilder⁴⁰.
- **Model creation.** Comprehensive modeling capabilities with multiple physics, programming languages, and high-fidelity simulations are essential. For instance, TwinBuilder⁴⁰ and MindSphere⁴⁴ enable the creation of **multidisciplinary** domain models, including structure, fluid, and electromagnetic.
- **Model conversion.** It must possess the capability to perform format conversion from various input formats to the designated target format.
- **Model lightweight.** Fast lightweighting of large-scale models should be supported.
- **Model assembly.** The ability to construct complex models from basic models is essential, such as using component models to build complex equipment model with multi-layer spatio-temporal correlations. For instance, Azure Digital Twins⁴² and 3DEXPERIENCE⁴³ support model assembly to a certain degree.
- **Model fusion.** Establishing coupling relationships between models in different disciplines and fields must be supported. For example, TwinBuilder⁴⁰ realizes the coupling of multi-physics field models.

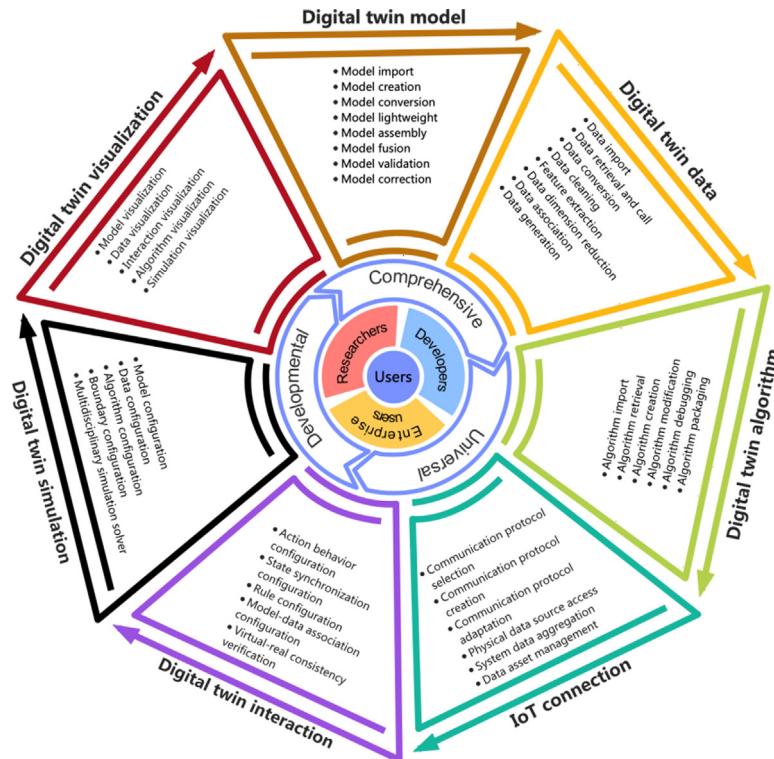


Fig. 1 Demands of users for digital twin software platform.

Table 1 Functional demands of users for digital twin software platform.

Core functions	Functional demands	Core functions	Functional demands
F₁: Digital twin model	F_{1.1}: Model import F_{1.2}: Model creation F_{1.3}: Model conversion F_{1.4}: Model lightweight F_{1.5}: Model assembly F_{1.6}: Model fusion F_{1.7}: Model validation F_{1.8}: Model correction	F₅: Digital twin interaction	F_{5.1}: Action behavior configuration F_{5.2}: State synchronization configuration F_{5.3}: Rule configuration F_{5.4}: Model-data association configuration F_{5.5}: Virtual-real consistency verification
F₂: Digital twin data	F_{2.1}: Data import F_{2.2}: Data retrieval and call F_{2.3}: Data conversion F_{2.4}: Data cleaning F_{2.5}: Feature extraction F_{2.6}: Data dimension reduction F_{2.7}: Data association F_{2.8}: Data generation	F₆: Digital twin Simulation	F_{6.1}: Model configuration F_{6.2}: Data configuration F_{6.3}: Algorithm configuration F_{6.4}: Boundary configuration F_{6.5}: Multidisciplinary simulation solver
F₃: Digital twin algorithm	F_{3.1}: Algorithm import F_{3.2}: Algorithm retrieval F_{3.3}: Algorithm creation F_{3.4}: Algorithm modification F_{3.5}: Algorithm debugging F_{3.6}: Algorithm packaging	F₇: Digital twin visualization	F_{7.1}: Model visualization F_{7.2}: Data visualization F_{7.3}: Interaction visualization F_{7.4}: Algorithm visualization F_{7.5}: Simulation visualization
F₄: IoT Connection	F_{4.1}: Communication protocol selection F_{4.2}: Communication protocol creation F_{4.3}: Communication protocol adaptation F_{4.4}: Physical data source access F_{4.5}: System data aggregation F_{4.6}: Data asset management.		

- **Model validation.** The ability to proficiently validate the precision of the developed digital twin model is essential. For example, TwinBuilder⁴⁰ and Azure Digital Twins⁴² provide the ability to verify and optimize twin models.
 - **Model correction.** It must possess the capability to execute model corrections. For example, the Core Simulation Live⁴⁵ enables users to update analysis results while modifying geometric entities dynamically.
- (2) Digital twin data

When carrying out digital twin data research and application development, users expect to be able to quickly and adeptly process heterogeneous data from multiple sources while meeting the following requirements.

- **Data import.** First, users hope to import various types and formats of data and achieve format compatibility with heterogeneous data. For instance, 3DEXPERIENCE⁴³ is capable of directly importing up to 82 data formats, while AWS IoT TwinMaker⁴¹ also support the importation of multi-source and heterogeneous data.
- **Data retrieval and call.** Ensuring data integrity is essential while retrieving and calling data. For example, 3DEXPERIENCE⁴³ solves various problems such as data standardization, low efficiency of data design tools, and poor accuracy of manual data, enabling the automation of processing complex and diverse full-space 3D geographic information spatial data in a simple and fast way.

- **Data cleaning.** Integrating, filtering, correlating, and synthesizing data is necessary to facilitate judgment, planning, validation, and diagnosis. For example, Beacon⁴⁶ and Predix⁴⁷ support the fusion of multi-source and heterogeneous data and virtual data collected. Meanwhile, with the help of data fusion software Pycharm, the efficiency of debugging, syntax highlighting, project management, code jumping, intelligent prompts, automatic completion, unit testing, and version control can be improved.
 - **Data conversion.** It should be able to transmit and exchange data through one or more data links.
 - **Feature extraction.** The function should identify the most effective features in the original features and convert them into features with clear physical meaning, geometric features, or statistical significance.
 - **Data association.** It should be possible to associate uncertain multi-source and heterogeneous data observations with trajectories.
 - **Data dimension reduction.** It should be capable of performing dimensionality reduction on both linear and non-linear data.
 - **Data generation.** It refers to the ability to generate virtual data with a similar distribution to the actual data in specific operating conditions with simulation. For example, Unity3D Computer Vision Datasets⁴⁸ solves the issue of insufficient training data when developing computer vision applications.
- (3) Digital twin algorithm

Users expect the digital twin algorithm to implement built-in or custom algorithm models with the following capabilities.

- **Algorithm import.** It requires to support algorithm models to be developed in multiple languages, such as Python, C++, Java, etc. For example, Azure Digital Twins⁴² supports running Kusto queries and advanced algorithms using Python plugin.
- **Algorithm retrieval.** It needs to retrieve and call multiple algorithm models from the algorithm library.
- **Algorithm creation.** Typically, users aspire to generate customized algorithms utilizing the development environment's assistance. Software with this capability includes Unity Manufacturing Toolkits,⁴⁹ Predix,⁴⁷ etc.
- **Algorithm modification.** With an editable interface of the algorithm, users are allowed to modify the algorithm structure and parameters.
- **Algorithm debugging.** It requires to support the compilation and debugging of the algorithm model.
- **Algorithm packaging.** It refers to combining algorithm models' properties, structures, and operations into a standalone entity.

(4) IoT connection

When carrying out an IoT connection, that is, realizing the virtual-real interconnection of digital twin, users anticipate the following requirements to facilitate the establishment of a data bridge between the physical space and virtual space.

- **Communication protocol selection.** A rich set of encapsulated communication protocols are required to be called upon. For example, Unity Manufacturing Toolkits⁴⁹ and Predix⁴⁷ incorporate standard module libraries for mainstream communication protocols.
- **Communication protocol creation.** It should be possible to create communication protocols for different physical devices.
- **Communication protocol adaptation.** For instance, Predix⁴⁷ supports the connection of physical devices through multiple industrial protocols via configuration. AWS IoT TwinMaker⁴¹ enables IoT connectivity for different devices through communication protocol configuration. Emulate3D⁵⁰ supports multiple communication protocols such as PCC, Profibus, OPC, Modbus and CANopen, and can configure communication protocols for various PLC brands.
- **Physical data source access.** Real-time data collection can be achieved by connecting physical products/equipment through IoT. For example, Thingworx⁴⁵ can connect various smart devices to the IoT ecosystem for data collection. Software tools with similar functions include MindSphere⁴⁴ and Azure Digital Twins⁴² built-in connector supports connection to physical space devices through IIoT, and AWS IoT TwinMaker⁴¹ can be used to collect and analyze data on large-scale industrial equipment.
- **System data aggregation.** It's crucial to enable the digital twin system to obtain data from multiple information systems. For instance, the AWS IoT TwinMaker⁴¹ Data Connector connects to AWS IoT SiteWise for asset model data and to Amazon S3 for resources like visual asset files and

device documentation. TwinBuilder⁴⁰ incorporates a built-in connector to integrate the packaging SDK with the IIoT and send and receive operational data.

- **Data asset management.** It is necessary to support planning, controlling, developing, executing, and supervising data and information assets for the digital twin system.
- #### (5) Virtual-reality interaction

When conducting virtual-reality interaction for digital twin, it is necessary to adhere to the following requirements to achieve synchronous interaction and feedback control between the virtual and physical actions.

- **Action behavior configuration.** The ability to synchronize the action of physical products/equipment with the virtual model is inevitable. For instance, the Unity Manufacturing Toolkit⁴⁹ realizes synchronization between virtual and real actions through the physical state data collected in real-time after configuring action constraints related to equipment/products. Azure Digital Twins⁴² also has similar functionality.
- **State synchronization configuration.** The states consistency between virtual and real space should be guaranteed. For example, 3DEXPERIENCE⁴³ can maintain consistency between real and virtual assembly actions and states by connecting physical space data, and TwinBuilder⁴⁰ can simulate synchronization between real and virtual by collecting physical product state data in real-time.
- **Rule configuration.** It requires providing action and state rule configuration.
- **Model-data association configuration.** It aims to realize the interactive relationship between the model and the data in virtual-reality interaction.
- **Virtual-real consistency verification.** It needs to verify the consistency of the actions and states of the digital twin models and the physical entities.

(6) Virtual simulation

Users anticipate being able to conduct virtual simulations using the developed digital twin models, data, algorithms, and interactions among other components, to accurately replicate the properties and parameters of the physical space. The capability of the virtual simulation function should meet the following requirements.

- **Model configuration.** Before the virtual simulation, the relevant parameters of the digital twin models can be configured.
- **Algorithm configuration.** Similarly, it needs to support algorithm parameters configuration.
- **Data configuration.** It needs to support configuration of real-time and historical data.
- **Boundary configuration.** It is necessary to define the parameters that establish the boundaries of the simulation process.
- **Multidisciplinary simulation solver.** It is important that the model from different disciplines and domains should be solvable. For example, MindSphere⁴⁴ provides the function of modeling and simulating complex systems in a multidisciplinary domain for digital twin. TwinBuilder⁴⁰ supports

the integration and co-simulation of multi-physical domain system models in various ways. 3DEXPERIENCE⁴³ facilitates electromagnetic, material science, fluid, structure, and multi-body simulation. Emulate3D⁵⁰ offers support for the process simulation of complex automation systems. MIIoT.VC⁵¹ is employed for digital industrial simulation, such as 3D process simulation, assembly simulation, logistics simulation, robot simulation, and virtual commissioning. Altair Activate⁵² combines physical and data-driven technologies to help users understand the overall performance of the system and identify the interrelationships between subsystems in the early stages of product development. Emulate3D⁵⁰ can achieve virtual simulation and debugging of the entire factory automation control system.

(7) Visualization

When conducting digital twin research and application, users anticipate a digital twin software platform that offers enhanced visual representations for digital twin models, data, algorithms, connectivity, interactions, and simulations.

- **Model visualization.** It provides visualization for the movement, position, and state of a 3D geometry model. For example, ANSYS supports embedded software integration with SCADE Display⁵³ for model visualization. ThingWorx⁴⁵ enables augmented reality bird's eye view to identify problems in a plant.
- **Data visualization.** Visualization for various data should be utilized to gain insight into problems and aid decision-making. For instance, AWS IoT TwinMaker⁴¹ provides alerts based on data visualization to determine when data or forecasts are out of expectations. Predix⁴⁷ can be employed for data analysis and monitoring and offers asset performance management and operational optimization services. ThingWorx⁴⁵ monitors and alerts on equipment status with real-time data collection.
- **Interaction visualization.** Visualization for virtual-reality interaction is required. For example, 3DEXPERIENCE,⁴³ TwinBuilder,⁴⁰ and Simcenter⁴⁴ all support virtual-real interaction visualization.
- **Algorithm visualization.** It aims to provide visualization of the solving process and results of the algorithm.
- **Simulation visualization.** It needs to offer a visual representation of the process and results of diverse simulation models. For instance, 3DEXPERIENCE⁴³ uses the SIMULA software set to build a 3D simulation ecosystem and visualize the simulation. TwinBuilder⁴⁰ supports the visualization of multidisciplinary and multi-physics simulation processes.

As a fusion of fundamental scientific and engineering principles, industrial software integrates industrial technology with information technology. The successful execution of digital twin application is contingent upon the utilization of industrial software. Nowadays, numerous renowned software suppliers have developed digital twin software platforms based on their extensive technical expertise and business requirements. Nevertheless, due to factors such as digital twin insights, business domain scope, and their own technology, the digital twin software platforms in the market have significant limitations in terms of functionality.

(1) For researchers, the existing digital twin industrial software platforms are deemed inadequate as they are mainly

designed for conventional algorithm testing, multi-physical finite element simulation, or general data storage and analysis. These platforms are often inclined towards specific domains, such as geometry modeling, mechanism analysis, or IoT platforms. Furthermore, there is a lack of a system tool platform that caters specifically to digital twin theory research, including the fusion of geometry-physics-behavior-rule for digital twin models, acquisition-transmission-processing-use for digital twin data, and complementary-iteration-coevolution for both digital twin model and data.

(2) For developers, in the realm of software development for digital twin, existing industrial platforms typically offer only a subset of the core functionalities, such as 3D visualization and interaction, finite element analysis, multi-physics field calculations, or machine learning algorithm design. To cater to distinct and customized digital twin functional specifications, it is often necessary to manually integrate multiple development and compatible tools for individual projects. At present, there's still a lack of a universal software development platform that is equipped with robust open-source, all-encompassing features and superior compatibility.

(3) For enterprise users, the current digital twin industrial software platforms are highly specialized, providing professional-grade multi-physical coupling analysis, diverse algorithms, and advanced 3D visualization development techniques. However, there is a significant challenge for common enterprise users who struggle to customize digital twin application service systems to align with their specific requirements. Consequently, a configurable, modular, and universally applicable digital twin development platform that can support the creation and implementation of digital twin application services is urgently required.

3. makeTwin: Platform architecture and core functions

Based on the deep exploration of digital twin theory and application, a software platform, makeTwin, is designed to meet the urgent demand for a universal digital twin software platform and to promote the implementation of digital twin theory. This section will focus on its design concept and ten core functions.

3.1. From the five-dimensional digital twin model to makeTwin

In the past few years, digital twin theory has gained significant attention from academia and industry, and the exploration of digital twin applications in various fields has entered a critical stage. Based on Grieves' three-dimensional digital twin model³, a five-dimensional model was proposed in 2017, which includes physical entity, virtual model, digital twin data, connection, and service.²² Subsequently, the theory system of digital twin model,¹¹ data,¹² connection and interaction^{13–14} have been successively proposed based on the five-dimensional model. Enabling technologies and tools for digital twin construction have also been explored to provide clear guidance for the digital twin construction method.¹⁵ Moreover, a set of digital twin standard systems has been jointly established with standardization administration and application enterprises, focusing on digital twin understanding and communication, the research and implementation of key technologies, and industry applications.²⁴ With the increasing improvement of digital twin theory, scholars and engineers from various industries realize

that digital twin applications should be based on industrial software to fully realize their industry value.

Under the guidance of the research on digital twin basic theories and common key technologies mentioned above, and inspired by the concept of “Make More Digital Twins”,¹ a general, comprehensive, open, and scalable industrial software platform reference architecture, makeTwin, has been developed. The original design architecture and implementation have been worked out.⁵⁴ This platform aims to provide a research and development reference architecture for digital twin holistic construction and service deployment for different users and application scenarios.

3.2. makeTwin: Platform architecture

The architecture of makeTwin platform is comprised of ten fundamental functional modules, namely *twinModelBuilder*, *twinDataProcessor*, *twinAlgBuilder*, *twinIoTConnector*, *twinInteractor*, *twinSimulator*, *twinLibrary*, *twinVisualization*, *twinSceneTemplate*, and *twinAppDeployer*, as shown in Fig. 2. Each module provides interoperability with features such as configurability, callability, modifiability, and extendibility. The eight core functions of makeTwin (excluding *twinAppDeployer* and *twinLibrary*) can interact through customizable API. makeTwin aims to provide a public platform for digital twin theoretical innovation research and verification for academic researchers, offer a general digital twin configurable and reusable development platform for project developers, and supply convenient and reliable modular digital twin services for enterprise users.

3.3. makeTwin: Ten core functions

3.3.1. twinModelBuilder

The primary function of *twinModelBuilder* is to manage digital twin models. To achieve multidimensional, multiscale, and multidomain descriptions and characterizations of physical entities or complex systems, *twinModelBuilder* adheres to the theoretical system of digital twin model construction¹¹ and consists of two functional sub-modules, which are *ModelContainer* and *ModelIntegration*.

The *ModelContainer* comprises four essential functions, which are model import, model creation, model conversion, and model lightweight. Their specific features are described as follows.

- **Model import.** The model import function enables the importation of geometry models in diverse formats and is compatible with mechanism models (physics, behavior, rule models) developed by different languages.
- **Model creation.** The model creation function supports the creation of multidomain (thermodynamics, mechanics, electromagnetics, etc.) and multidimensional (physics, behavior, rule, etc.) mechanism models.
- **Model conversion.** The model conversion function facilitates the format conversion of various geometry models and is compatible with physics, behavior, and rule models developed by different languages.
- **Model lightweight.** The model lightweight function allows for removing non-essential features of 3D geometry models at varying levels of granularity.

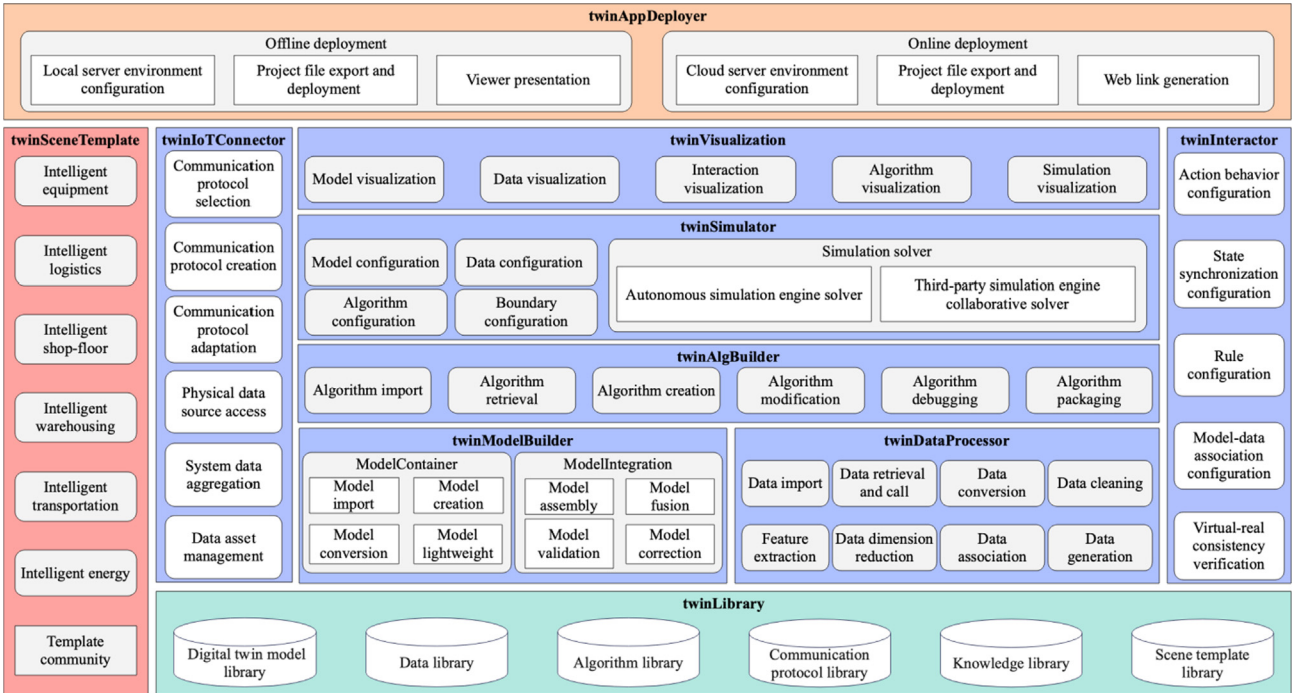


Fig. 2 Architecture for makeTwin software platform.

The ModelIntegration contains four additional functions, which are model assembly, model fusion, model validation, and model correction. Their specific features are described as follows.

- **Model assembly.** The model assembly function involves the process of assembling individual models, unit-level models, system-level models, and complex-system-level models in the spatial dimension. This process includes the creation of topological and constraint relationships.
- **Model fusion.** The model fusion function establishes coupling relationships between different disciplinary and domain models, mainly involving coupling relationship selection and model fusion.
- **Model validation.** The model validation function entails verifying the consistency between digital model output and physical object output based on various requirements, including individual model validation and fusion model validation.
- **Model correction.** The model correction function denotes the process of adjusting model parameters to rectify deviations. This primarily involves selecting correction parameters, rectifying deviations, and adjusting parameters.

3.3.2. twinDataProcessor

twinDataProcessor has been designed to equip makeTwin with advanced digital twin data management capabilities. To meet the criteria of digital twin data construction and processing, including complementary, standardization, timeliness, association, fusion, information growth, and service,¹² the twinDataProcessor module provides various features as follows.

- **Data import.** The data import function enables makeTwin to import various types of historical data of users, with compatibility for different mainstream data formats.
- **Data retrieval and call.** The function of data retrieval and call allows users to search and access specific data quickly.
- **Data cleaning.** The data cleaning function comprises data review and verification, deduplication, and error correction.
- **Data conversion.** The data conversion function supports the conversion of data from various sources into uniform formats.
- **Feature extraction.** The feature extraction function provides resources for users to extract key features from digital twin data, including causality, similarity, complementarity, etc.
- **Data association.** The data association function supports integrating data from different sources or databases into a unified database through shared fields or variables, enabling cross-database querying and data usage across different databases.
- **Data dimension reduction.** Data dimension reduction supports dimensionality reduction for high-dimensional data.
- **Data generation.** Data generation allows data augmentation through digital twin mechanism model simulation, digital twin data deduction, machine learning generation, etc.

3.3.3. twinAlgBuilder

The primary objective of twinAlgBuilder is to equip makeTwin with algorithm model development and management

capabilities. To meet algorithm models' configurability, verifiability, and optimization requirements, twinAlgBuilder offers various features, including multi-language algorithm import, algorithm call, algorithm creation, algorithm modification, and algorithm debugging. The specific functionalities of these features are described as follows.

- **Algorithm import.** The algorithm import function allows for the import of algorithm models developed by mainstream third-party languages such as Python, MATLAB, C++ , etc.
- **Algorithm retrieval.** The algorithm retrieval function supports the compatibility and retrieval of existing algorithm models in the algorithm library.
- **Algorithm creation.** The algorithm creation function empowers users to develop custom algorithm functions using the built-in Python editor.
- **Algorithm modification.** The algorithm modification function enables users to call existing algorithms from the library and modify algorithm structures and parameters using the algorithm function editor interface.
- **Algorithm debugging.** The algorithm debugging function is used for algorithm running and debugging, providing performance optimization related to model correctness, robustness, time complexity, space complexity, and other factors.
- **Algorithm packaging.** The algorithm packaging function is responsible for encapsulating the algorithm code, dependencies, and necessary data into a deployable unit that can be accessed through a set of API keys. Once encapsulated, other applications or systems can use the algorithm model without requiring knowledge of the underlying code or dependencies.

3.3.4. twinIoTConnector

twinIoTConnector plays a critical role in connecting the virtual and physical worlds. To meet the digital twin guidelines of accuracy, real-time, consistency, security, and reliability,¹³ twinIoTConnector sets up the bridge to achieve virtual-reality interaction, with features such as communication protocol selection, communication protocol creation, communication protocol configuration, physical data source access, system data aggregation, and data asset management. Their specific functionalities are described as follows.

- **Communication protocol selection.** The communication protocol call function allows users to call and use existing protocols from the communication protocols library.
- **Communication protocol creation.** The communication protocol creation function authorizes users to customize communication protocols based on protocol statutes.
- **Communication protocol adaptation.** The communication protocol configuration function is responsible for configuring communication protocols for different physical devices.
- **Physical data source access.** The physical data source access function collects and aggregates heterogeneous physical data from multiple sources of various physical resources.
- **System data aggregation.** The system data aggregation function can access and collect enterprise information system data such as MES, ERP, WMS, etc.

- **Data asset management.** The data asset management function can assist the equipment management and visual statistical analysis of data assets.

3.3.5. *twinInteractor*

The core function of the *twinInteractor* module is to provide *makeTwin* with the capability of virtual-real interaction. The *twinIoTConnector* and *twinInteractor* collaborate to establish a bridge for virtual-real information exchange and sharing. They also deliver “data nutrients” for dynamic updates of digital twin models, real-time control of physical entities, and online optimization of decision-making solutions to accelerate information sharing and integration. Its critical features include action behavior configuration, state synchronization configuration, rule configuration, digital model association configuration, virtual-real consistency verification, etc. Their specific functionalities are described as follows.

- **Action behavior configuration.** The action behavior configuration function allows the configuration of action form, motion path, and motion coupling relationship for the 3D geometry model in the virtual space so as to realize the motion synchronization between the virtual and physical spaces.
- **State synchronization configuration.** The state synchronization configuration enables the configuration and updating of the operation state of physical resources into the virtual space, aiming to achieve state synchronization of the virtual and physical spaces.
- **Rule configuration.** This function module is designed to configure relevant rules and constraints for the geometry models’ motion range, safety boundary, and interruption conditions in the virtual space.
- **Model-data association configuration.** The model-data association configuration function facilitates the association configuration between the digital twin model and enabling signals in virtual space.
- **Virtual-real consistency verification.** The virtual-real consistency verification function can be used to verify whether the virtual and reality actions or states are consistent with each other.

3.3.6. *twinSimulator*

The primary function of *twinSimulator* is to equip *makeTwin* with high-precision computing and simulation capabilities. The *twinSimulator* module offers two types of simulation engines, which are third-party and customized. They can perform multidisciplinary, multi-physics, multi-scale, and multi-probability simulations tailored to different application scenarios. The key features of the *twinSimulator* include model configuration, algorithm configuration, data configuration, and boundary configuration. Once these configurations are set up, a simulation can be conducted. Customized simulation engines are available for virtual simulations, such as fault diagnosis and prediction, scheduling and optimization, and path planning. For complex simulation problems that the existing simulation engine cannot solve, *makeTwin* incorporates third-party simulation engines to provide joint simulation

solutions. In the case of multidisciplinary coupling simulation calculations, the ANSYS simulation interface can be employed for joint solution calculation. Additionally, Python development and debugging tools can be integrated and invoked by the *makeTwin* platform to run and solve Python-related algorithms.

3.3.7. *twinVisualization*

The core function of *twinVisualization* is to provide *makeTwin* 2D/3D visualization capabilities. The *twinVisualization* module includes five functions: model visualization, data visualization, interaction visualization, algorithm visualization, and simulation visualization.

- **Model visualization.** The model visualization function demonstrates the imported 3D models in the form of a graphical display. It is mainly used for model rendering, model static visualization, model information visualization, and model action visualization.
- **Data visualization.** The data visualization function converts the provided information or data into a visual context to make it easier and faster to understand. The visual context includes but is not limited to bar charts, line charts, scatter charts, pie charts, area charts, dashboard charts, Gantt charts, and tables.
- **Interaction visualization.** The interactive visualization function manipulates and understands data/models through dialogue and interaction between the human operator and the system. The interaction includes but is not limited to dialog boxes, checkboxes, buttons, sliders, knobs, voice, and gestures.
- **Algorithm visualization.** The algorithm visualization function describes complex algorithm models using graphical visualization. It includes but is not limited to visualization of algorithm structure, parameter debugging, execution process, and execution results.
- **Simulation visualization.** The simulation visualization function develops simulations for different scenarios by calling 3D model visualization, data visualization, interaction visualization, and algorithm visualization.

3.3.8. *twinLibrary*

twinLibrary is responsible for providing a comprehensive support system for *makeTwin*. It consists of a digital twin model library, an algorithm library, a data library, a knowledge library, a communication protocol library, and a scene template library. These functions are described as follows.

- **Digital twin model library.** The digital twin model library stores the digital twin models imported and developed by users, comprising 3D geometry, behavior, physics, and rule models.
- **Algorithm library.** The algorithm library contains algorithm models developed by users. It provides access to built-in general-purpose encapsulated algorithm models, such as data processing class models and intelligent optimization class models.
- **Data library.** The data library preserves physical data collected from physical resources, virtual data generated from simulation, processed feature data sets, and deduced data.

- **Knowledge library.** The knowledge library keeps knowledge documents, such as process documents, fault maintenance documents, and expert experience documents.
- **Communication protocol library.** The communication protocol library houses standard communication protocols for equipment virtual-real connection in different application scenarios.
- **Scene template library.** The scene template library stores and provides access to customized application scenario templates developed by users, shortening the development cycle when creating new digital twin application services.

3.3.9. *twinSceneTemplate*

The core function of *twinSceneTemplate* is to provide a collection of common scenario templates across various fields. These templates can be imported at the beginning of project development to assist users in quickly navigating digital twin application services. Once development is completed, the developed scenario can also be saved to the scene template library for future use. Users are able to combine the functions of monitoring, simulation, prediction, evaluation, optimization, and management services freely with *twinSceneTemplate* in different application scenarios. Currently, *makeTwin* offers a range of service scenarios, such as smart equipment, smart shop-floor logistics, stereo warehouse, and smart energy. Additionally, users have the ability to upload both general-purpose and specialized scenario templates to the scenario template community for broader usage and accessibility. For instance, the smart equipment application scenario includes the model of general-purpose equipment, communication protocols, and various service components required by the scenario. Users can choose the relevant functions based on their specific needs.

3.3.10. *twinAppDeployer*

twinAppDeployer is mainly designed to facilitate flexible deployment of *makeTwin* application services through multiple modes, including offline and online deployment. Offline deployment is geared towards devices which lack Internet connectivity, whereby the digital twin application system is installed locally after configuring the environment on a designated computer or server. Conversely, online deployment is intended for devices lacking Internet connectivity, allowing users to upload digital twin application service files to a cloud server environment and obtain web links to access the application as web pages.

4. *makeTwin*: Workflow and interaction mechanism

The operation and application of the *makeTwin* software platform are particularly crucial. Therefore, this section gives the workflow and interaction mechanism of *makeTwin*.

4.1. *makeTwin*: Workflow

In the *makeTwin* software platform, researchers, developers, and enterprise users can choose two main development modes for the digital twin service program based on their application needs. The first mode is application service development based

on scenario templates, and the second is customized construction, as shown in Fig. 3. The specific workflow is as follows.

Step 1: New project creation.

Step 2: Determination of using existing scene template or not. In this step, users can determine whether to use the scene templates or not based on the availability and actual application needs.

Step 3: Scene configuration. If scene templates are available and applicable, users can configure digital twin models, data, communication protocols, and algorithm models by accessing these scene template library. Some specialized models, data, communication protocols, and algorithm models can be customized and constructed. If not, users need to create and configure the scene with the help of various libraries. Then, the simulation solver and visualization are configured.

Step 4: Service App generation and deployment. In this step, users generate a service App through the *makeTwin* software platform and deploy the App.

Step 5: End.

4.2. *makeTwin*: Interaction mechanism

The interaction mechanism of *makeTwin* is the input and output correlation between various core functions. In the *makeTwin* platform, the interaction mechanism between *twinModelBuilder*, *twinAlgBuilder*, *twinDataProcessor*, *twinIoTConnector*, *twinInteractor*, *twinSimulator*, *twinVisualization*, *twinLibrary*, and *twinSceneTemplate* is mainly considered, as shown in Fig. 4.

(1) The main carrier for *twinModelBuilder* to interact with other modules is the digital twin model. *twinModelBuilder* inputs the generated digital twin models from the third-party software with various formats, the existing model from *twinLibrary*, the algorithms generated by *twinAlgBuilder*, and the verification information from *twinSimulator*. It outputs the newly generated digital twin models for *twinInteractor*, *twinSimulator*, *twinVisualization*, *twinSceneTemplate* and *twinLibrary*.

(2) The main carrier for *twinDataProcessor* to interact with other modules is data. *twinDataProcessor* inputs the collected data from *twinIoTConnector*, the simulation data from *twinSimulator*, and the historical data available in *twinLibrary*. It outputs the processed data to *twinAlgBuilder*, *twinInteractor*, *twinSimulator*, *twinVisualization*, and *twinSceneTemplate*.

(3) The main carrier for *twinAlgBuilder* to interact with other modules is the algorithm. *twinAlgBuilder* inputs the algorithm models available from *twinLibrary*, the real-time data offered by *twinIoTConnector*, the processed data from *twinDataProcessor*, and the algorithm validation information from *twinSimulator*. It outputs the new algorithm models for *twinModelBuilder*, *twinSimulator*, *twinVisualization*, *twinSceneTemplate*, and *twinLibrary*.

(4) The main carriers for *twinIoTConnector* to interact with other modules are communication protocols and data. *twinIoTConnector* inputs physical device data and enterprise system data obtained through communication protocols. It outputs real-time data for *twinAlgBuilder*, *twinDataProcessor*, *twinSimulator*, *twinVisualization*, *twinLibrary*, and *twinSceneTemplate*.

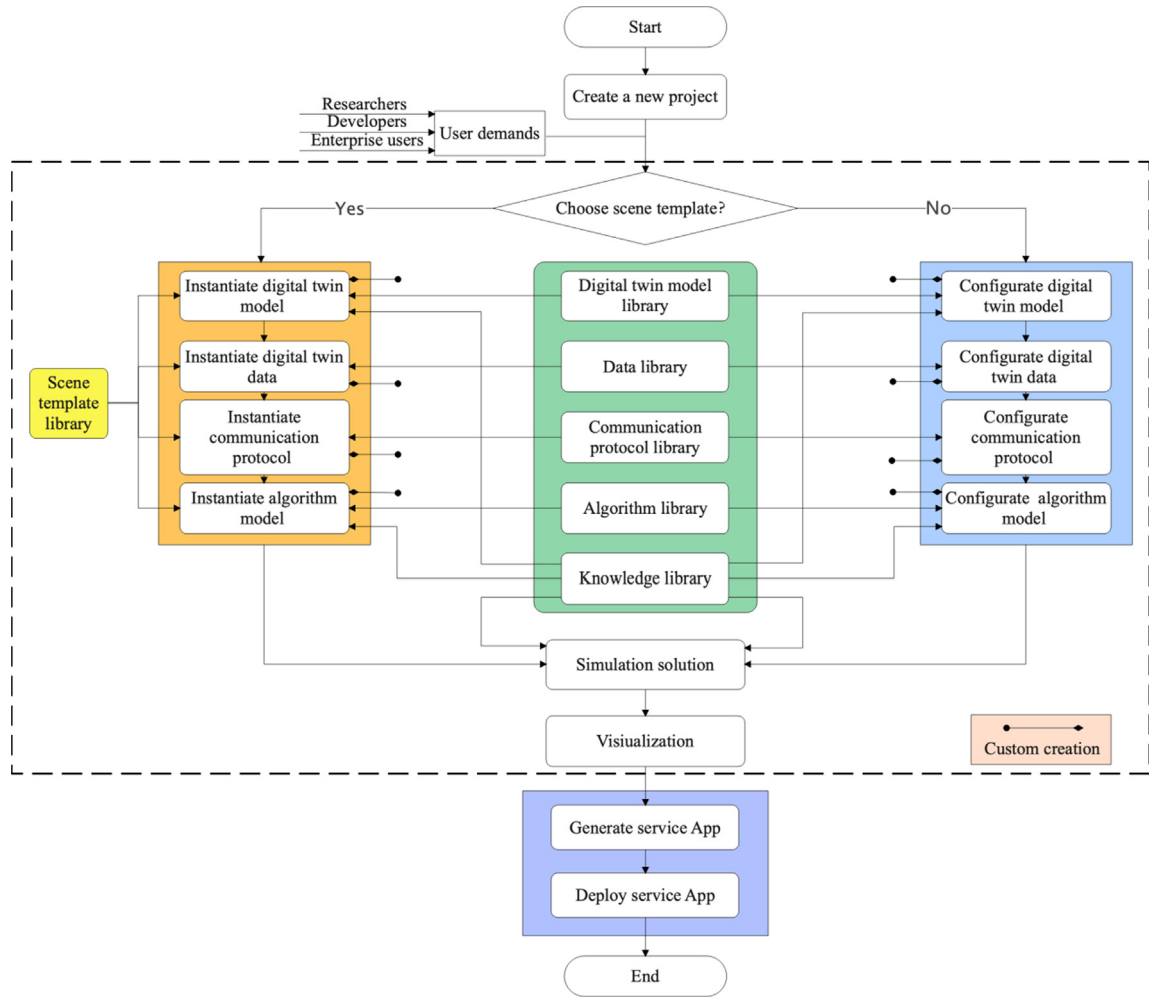


Fig. 3 Workflow for makeTwin.

(5) The main carriers for twinInteractor to interact with other modules are digital twin model and data. twinInteractor inputs digital twin models from twinModelBuilder, processed data from twinDataProcessor, and consistency validation information from twinSimulator. It outputs action and status synchronization configurations for twinSimulator, twinVisualization, and twinSceneTemplate.

(6) The main carriers for twinSimulator to interact with other modules are digital twin models, algorithm models, and data. twinSimulator inputs digital twin models from twinModelBuilder, algorithm models from twinAlgBuilder, processed data from twinDataProcessor, and action and status configuration information from twinInteractor. It outputs model validation information for twinModelBuilder, algorithm model validation information for twinAlgBuilder, simulation data for twinDataProcessor, consistency validation information for twinInteractor, and a simulation engine for twinVisualization and twinSceneTemplate.

(7) The main carriers for twinVisualization to interact with other modules are digital twin models, algorithm models, and data. twinVisualization inputs digital twin models, algorithm models, data, and simulation engines from various functional modules. And it outputs visualization components for twinSceneTemplate.

5. makeTwin: An industrial application and practice

In this section, a digital twin application system for a chemical fiber textile shop floor (CFTS), which was developed according to the proposed makeTwin reference architecture, is taken as an example to show the feasibility of the proposed reference architecture.

5.1. Brief introduction and problems description for the CFTS

As shown in Fig. 5, the CFTS includes a winding production unit, a silk ingot packaging unit, and a stereo warehouse storage unit. The winding production unit is responsible for the following listed tasks:

- Processing chemical raw materials into raw silk
- Winding raw silk into silk ingots
- Collecting and loading the silk ingots onto the staging shelf automatically

The silk ingot packaging unit is responsible for weighing, bagging, stacking, packaging, and film wrapping the silk ingots on the silk car. The stereo warehouse storage unit is responsi-

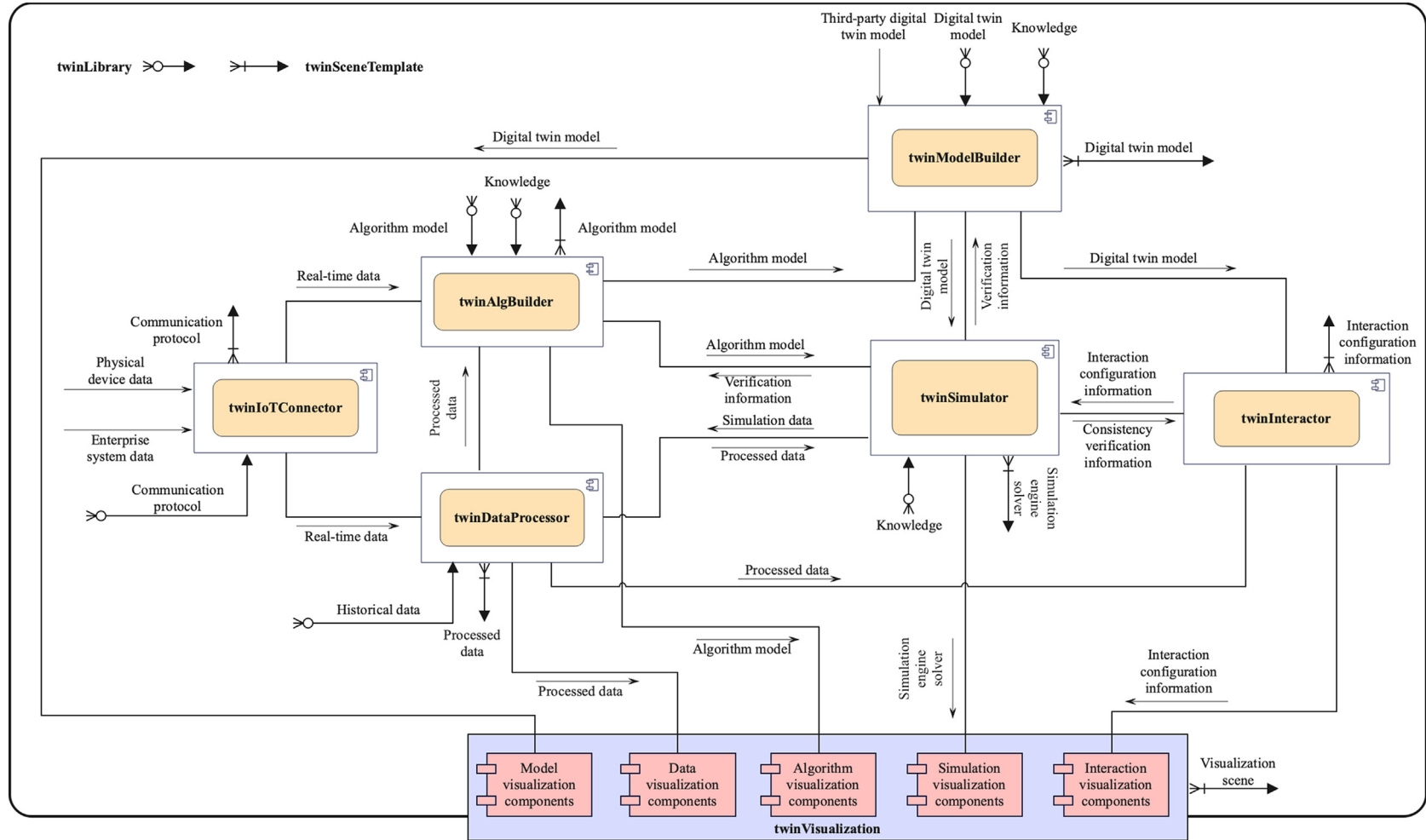


Fig. 4 Interaction mechanism for makeTwin.

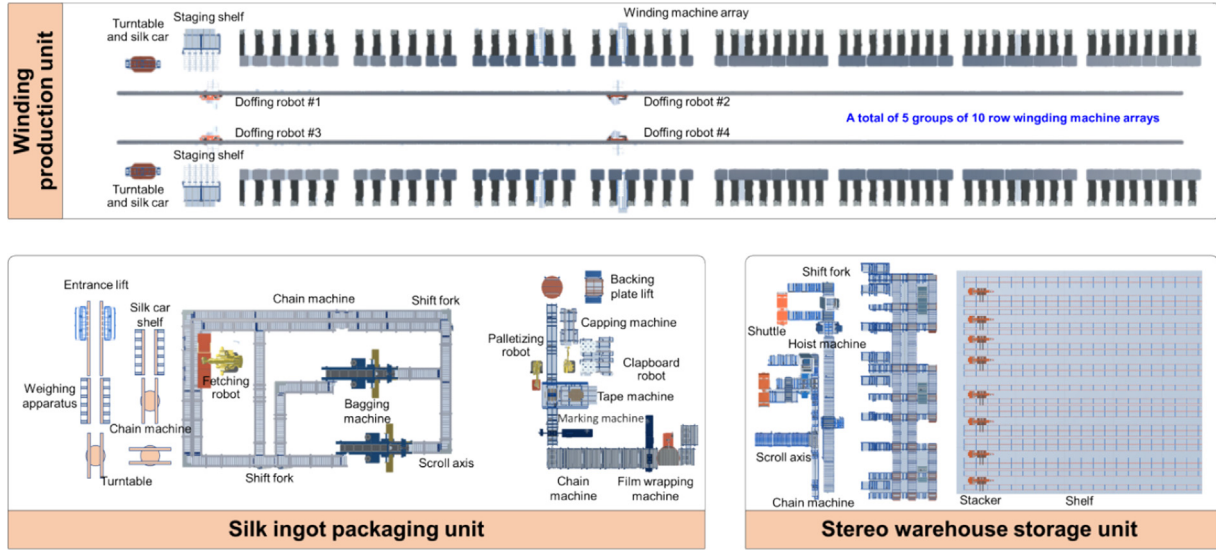


Fig. 5 Schematic diagram of the CFTS production line.

ble for storing the packaged goods and executing outbound operations on time.

The CFTS is renowned for its complex processes and numerous equipment. However, the using of previous information system on the shop-floor is constrained by its poor visibility, low job scheduling efficiency, and limited fault diagnosis accuracy. Specifically, the various equipment systems on the shop floor operate independently, making it difficult to establish correlation relationships between them to conduct a comprehensive analysis of the production process. Additionally, the existing visualization system is imperfect. Changes in the production process make the original system inadequate for current monitoring needs. Furthermore, the existing shop floor digital systems cannot configure production parameters and constraints, resulting in an unreasonable and inaccurate production plan simulation. Also, the current fault diagnosis strategies lack the ability to configure algorithm parameters, resulting in inefficient data processing, low-level data analysis, and an inability to determine the type and location of faults. Therefore, a digital twin system is urgently required for the CFTS.

5.2. Functional requirements of the CFTS digital twin system

To address the challenges mentioned above, three primary functional requirements have been identified for the CFTS digital twin system as follows.

(1) **Three-dimensional visualization with an intuitive experience.** Based on the real-time perception of production logistics data, the system should monitor the operating status of various textile equipment, display the production logistics process, detect fault information, and conduct data analysis.

(2) **Optimization of production logistics operations strategy.** The system should formulate production plans, allocate tasks automatically, and make real-time adjustments to production and logistics resources based on the shop floor's production capacity and production orders.

(3) **Diagnosis of textile equipment faults and predictive maintenance.** By monitoring the status and behavior of pro-

duction equipment (such as winding machines) and logistics equipment (such as creels, shuttle cars, and stackers) in real-time, the system should automatically detect, locate, and eliminate faults, as well as provide predictive maintenance strategies.

5.3. Development of the CFTS digital twin system

The CFTS digital twin system was designed and developed leveraging the proposed makeTwin architecture. Specifically, modules like *human-computer interaction interface*, *high-fidelity digital twin modeling*, *3D visual monitoring*, *shop-floor operation strategy optimization*, and *fault diagnosis and prognosis* are developed accordingly.

- 1) **Human-computer interaction interface.** As shown in Fig. 6, the main interface of the CFTS digital twin system includes a main menu bar and entrances to particular shop-floor scenes. The main menu bar includes *Home page*, *Scene selection*, *Fault diagnostics*, *Information tracing*, *Quality inspection*, *Virtual training*, *Job scheduling*, and *Remote collaboration*. The shop-floor scene entrances include the *Winding production unit*, *Silk ingot packaging unit*, and *Stereo warehouse unit*.
- 2) **High-fidelity digital twin modeling.** As shown in Fig. 7, geometry models are constructed to characterize the shapes, sizes, and relations of the equipment and production line. Physics models are created to characterize the stochastic degradation of the kinematic and mechanical performance of key components. Behavior models are developed to characterize the dynamic functions, job-shop rescheduling, and external disturbances of the shop floor. Rule models are also built to represent domain knowledge, production rules, logistics rules, and maintenance regulations. By integrating all the models above, a high-fidelity digital twin model is obtained for the accurate depiction of the physical characteristics of the CFTS.

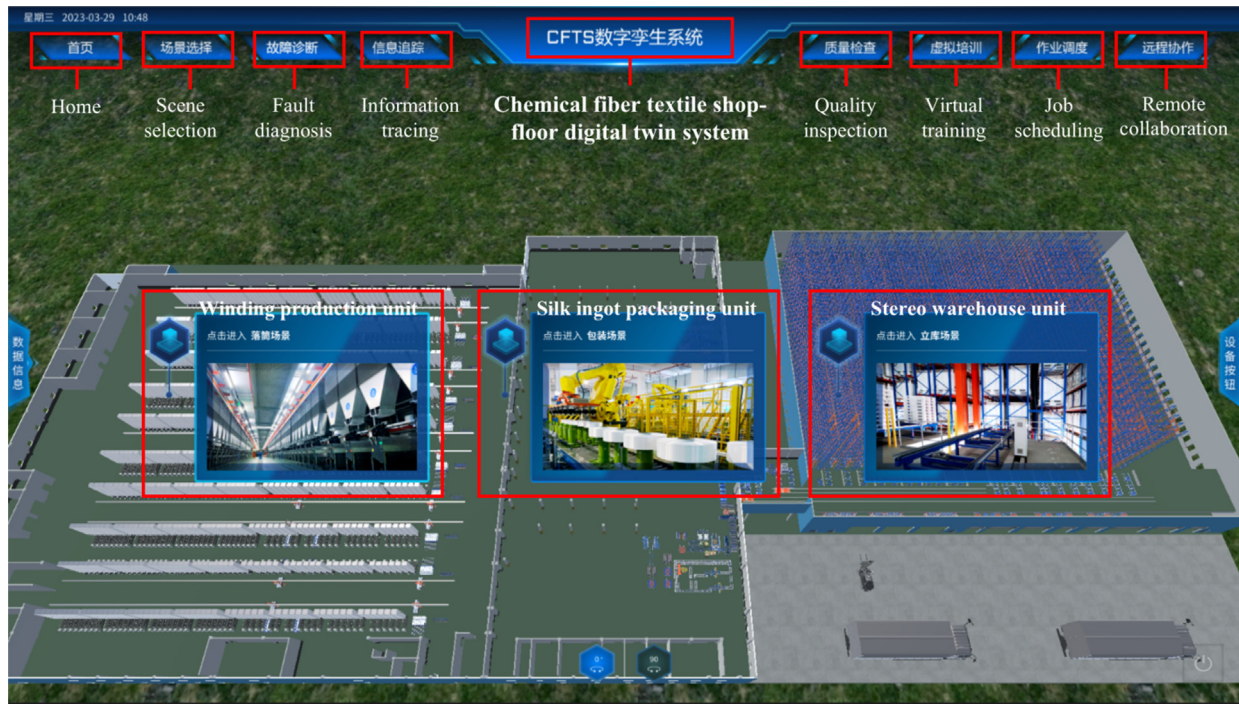


Fig. 6 The main interface of the CFTS digital twin system.

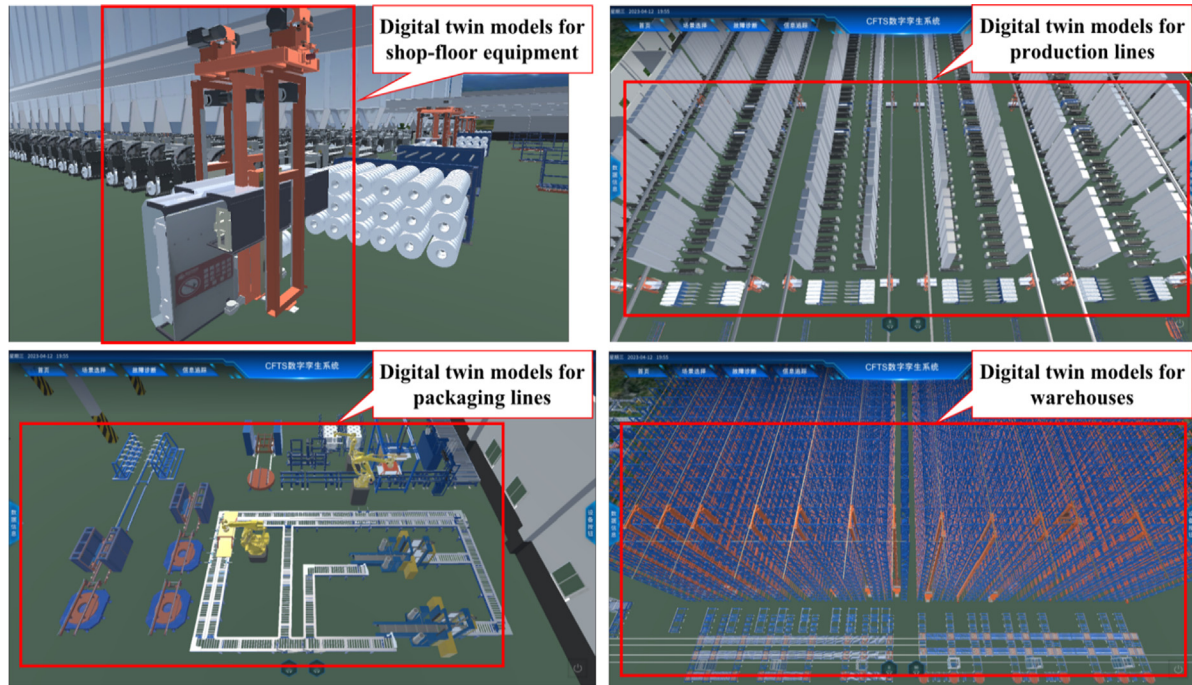


Fig. 7 High-fidelity digital twin modeling.

- 3) **3D visual monitoring.** As shown in Fig. 8, by building a data acquisition network on the shop floor, all data from the production and logistics system are collected in real-time at high frequency and stored in the database. The data includes production information (such as output, efficiency, progress, etc.), equipment status information (such as status, control, alarm, etc.), and

personnel operation information. Moreover, by correlating the acquired real-time data with the constructed shop-floor digital twin model, the virtual shop floor can be synchronized in real-time with the physical shop floor, ultimately achieving a three-dimensional visual monitoring of the CFTS with an intuitive experience.

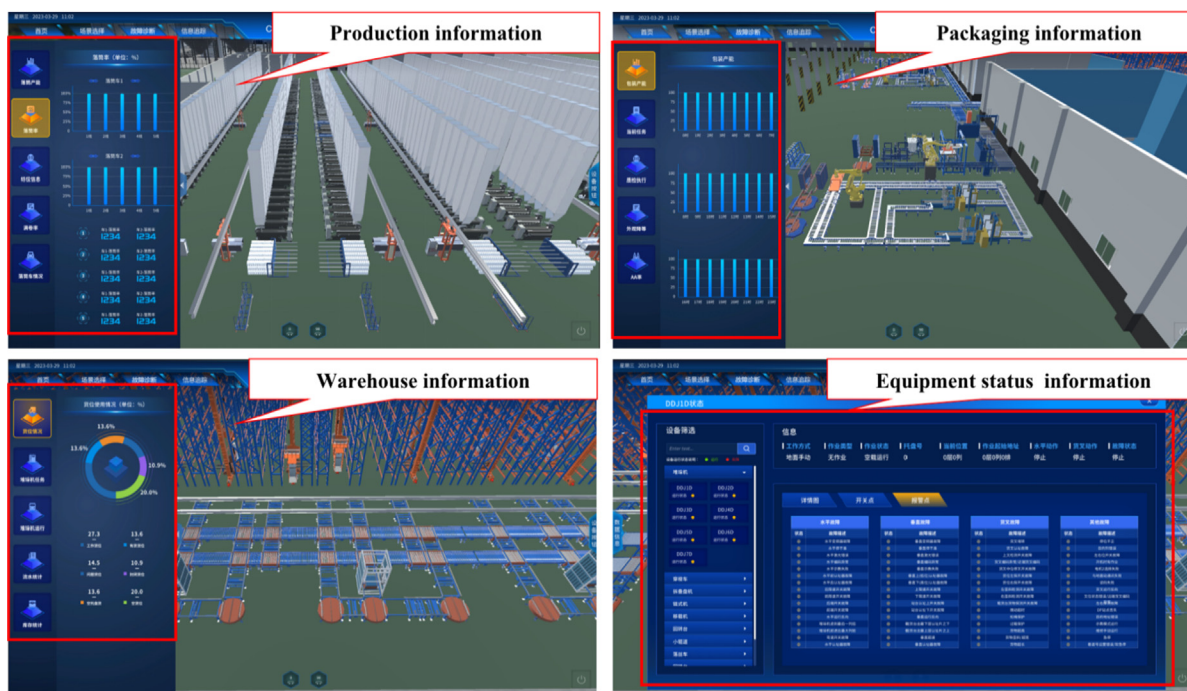


Fig. 8 3D visual monitoring.



Fig. 9 Optimization of shop-floor operation strategies.

- 4) **Shop-floor operation strategy optimization.** Efficient collaboration between production and logistics equipment, as well as effective job scheduling strategies, are critical to ensuring stable and efficient production and logistics in the CFTS. As shown in Fig. 9, adaptive algorithms for work time-span prediction, job-shop scheduling optimization, and warehouse space allocation have been developed. These algorithms support the enterprise to carry out virtual shop-floor operation testing before

actual production. With users' permission, it enables real-time adaptive scheduling optimization during actual operations, thereby improving production and logistics efficiency.

- 5) **Fault diagnostics and prognostics.** As illustrated in Fig. 10, a knowledge base is constructed for fault diagnostics, fault prognostics, fault reporting, and historical fault querying based on real-time/historical equipment failure and maintenance records, and also numbers of

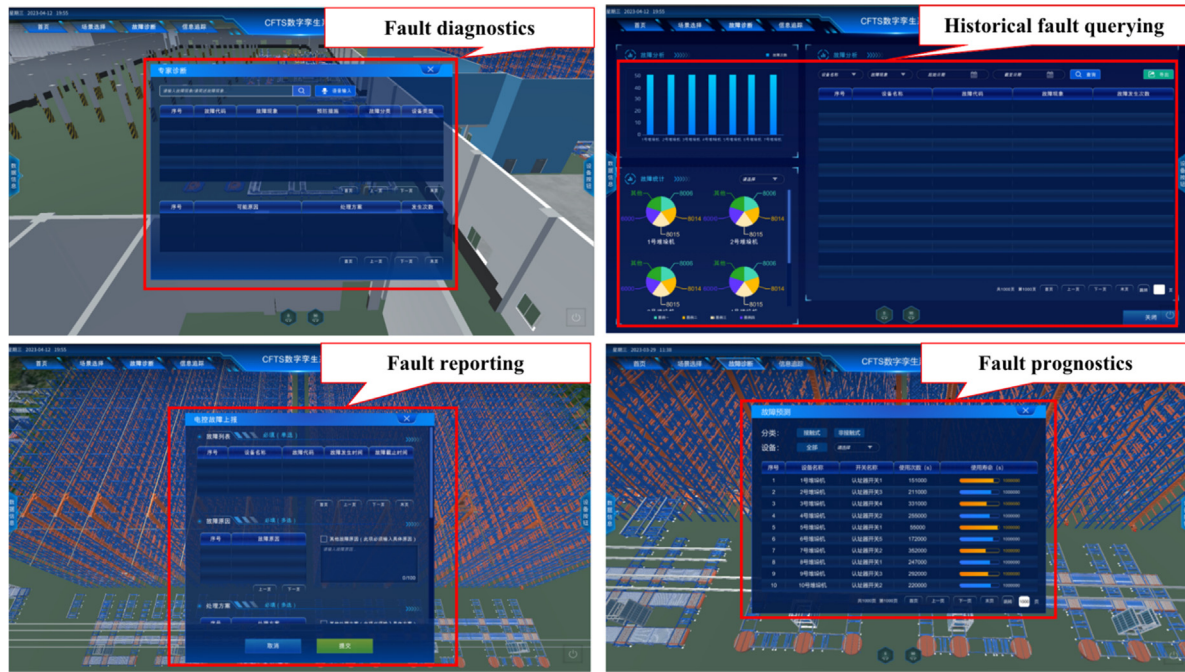


Fig. 10 Fault diagnosis and prognosis.

electric-hydraulic-energy-control coupling digital twin models is built, to forge a hybrid-driven method for the enhancement of equipment fault diagnosis and prognosis, which reduces equipment downtime, and thereby improves the production and logistics efficiency.

Overall, the CFTS digital twin system developed in this paper has achieved the following functions:

(1) Visual monitoring of the logistics and production line. Operators can view the operation status of the entire production line via the 3D visualization module, including equipment status, fault alarm information, production task information, and personal operation information.

(2) Adaptive scheduling operation optimization. It can provide scheduling optimization for the job scheduling of the winding, packaging, and warehousing units, as well as the stereo warehouse.

(3) Fault diagnosis and prognosis of key equipment components. For instance, it can effectively evaluate the wear and tear of vulnerable parts such as the running wheel and wire rope of the stacker crane, predict possible downtime, and adjust predictive maintenance strategies.

Empowered by the developed digital twin system, the CFTS has effectively improved the intelligence and informatization level of the production line and equipment and reduced equipment fault handling time by about 30% and manual operation costs by about 20%.

6. Conclusions and future works

The proposed makeTwin, which designed with the comprehensive consideration of high openness, compatibility, interoperability, availability and extensibility, can not only benefits the academic researchers, but also the developers and enterprise users. It aims to provide a platform to fulfill the contin-

uous developing requirements of digital twin research, application development and management. Moreover, with its well considered architecture and functions, the barriers of knowledge gathering, function reusing and work collaboration among interested bodies of digital twin can be eliminated.

However, the proposed makeTwin is the first exploration for an universally software platform in the field of digital twin, and the makeTwin is still in its early stage from the concept to implementation from the mature engineering software view. The architecture still needs perfecting, and the functions of modules still require continuous enrichment and improvement. Future works are as follows.

(1) The architecture and functions need to be further iterated and improved to forming an applicable engineering software, as well as ensuring the adaptability to various industrial and application scenarios.

(2) makeTwin deriving forms for different using scenarios need to be developed to ensure the using efficiency for specific industry.

(3) More makeTwin industry applications will be carried out to verify and then to improve it.

(4) Related standards need to be developed to support and guide the research and industrial application of digital twin.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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